

FACIAL NERVE

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DIKIOHS,DUHS.

LEARNING OBJECTIVES

- At the end of the lecture ,students should be able to
- Describe functions of facial nerve
- Identify components of facial nerve
- Relate nerve to facial functions
- Explain Bell's Palsy.

YOUTUBE LINKS

Cranial Nerves: Functional Components

https://youtu.be/5L_7HVzHBko

Facial Nerve: Nuclei and Functional Components

<https://youtu.be/CmhZMHrtf5o>

Introduction

- The facial nerve is the seventh cranial nerve.
- Mixed nerve with motor and sensory roots.
- Emerges from the brainstem between the pons and medulla.
- It contains the motor, sensory, and parasympathetic (secretomotor) nerve fibers, which provide innervation to many areas of the head and neck region.
- The facial nerve is comprised of three nuclei:
 - The main motor nucleus
 - The parasympathetic nuclei
 - The sensory nucleus

- Carries parasympathetic secretory fibers to submandibular and sublingual salivary glands, lacrimal gland and to mucous membranes of oral and nasal cavities.
- Conveys exteroceptive sensation from eardrum and external auditory canal, proprioceptive sensation from muscles it supplies, and general visceral sensation from salivary glands and mucosa of the nose and pharynx.

Structure and Function

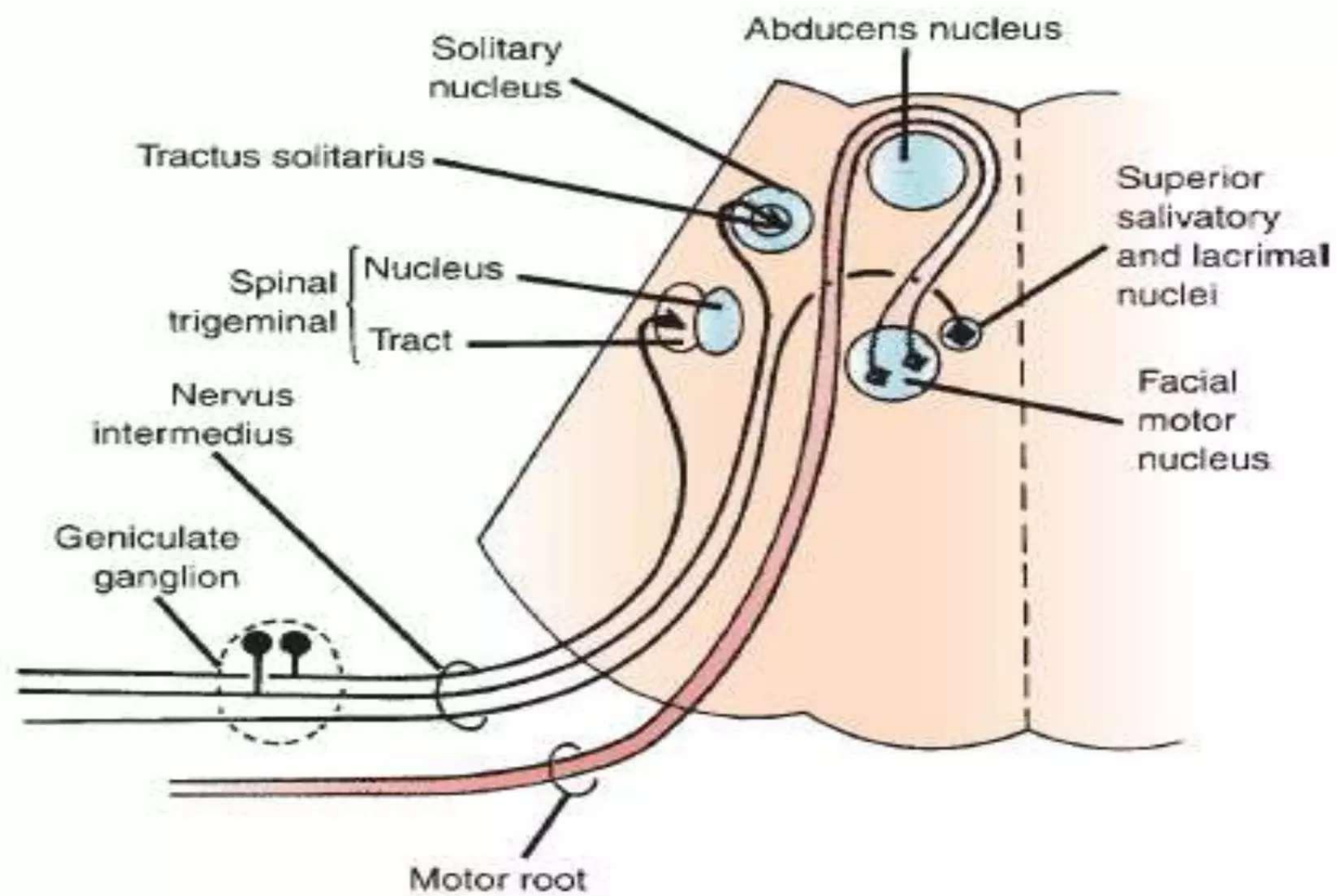
- There are four major functions of the facial nerve:
- General somatic efferent (motor supply to facial muscles)
- General visceral efferent (parasympathetic secretomotor supply to submandibular and sublingual salivary glands and the lacrimal gland)
- Special visceral afferent (taste sensation from the anterior two-thirds of the tongue)
- General somatic afferent (cutaneous sensations from the pinna and the external auditory meatus)

Facial nerve is formed mainly of two parts:

➤ **1 Facial nerve proper (motor):**

- Arising from facial motor nucleus in pons.
- Supranuclear innervation to the muscles of facial expression arises from the lower third of contralateral precentral gyrus in facial area of motor homunculus.
- Portion of nucleus that innervates the lower half to two thirds of the face has predominantly contralateral supranuclear control;
- Portion that innervates upper third to half has bilateral control.

- Facial nucleus is special visceral efferent, or branchiomotor;
- It innervates the muscles of the second branchial arch
- Facial motor nucleus has lateral, medial, and dorsal subnuclei, arranged in columns.
- Exits the pons laterally at the pontomedullar junction, just caudal to the roots of CN V between the olive and the inferior cerebellar peduncle
- Has two components, motor root, which makes up about 70% of the fibers, and sensory root, which accounts for 30%.



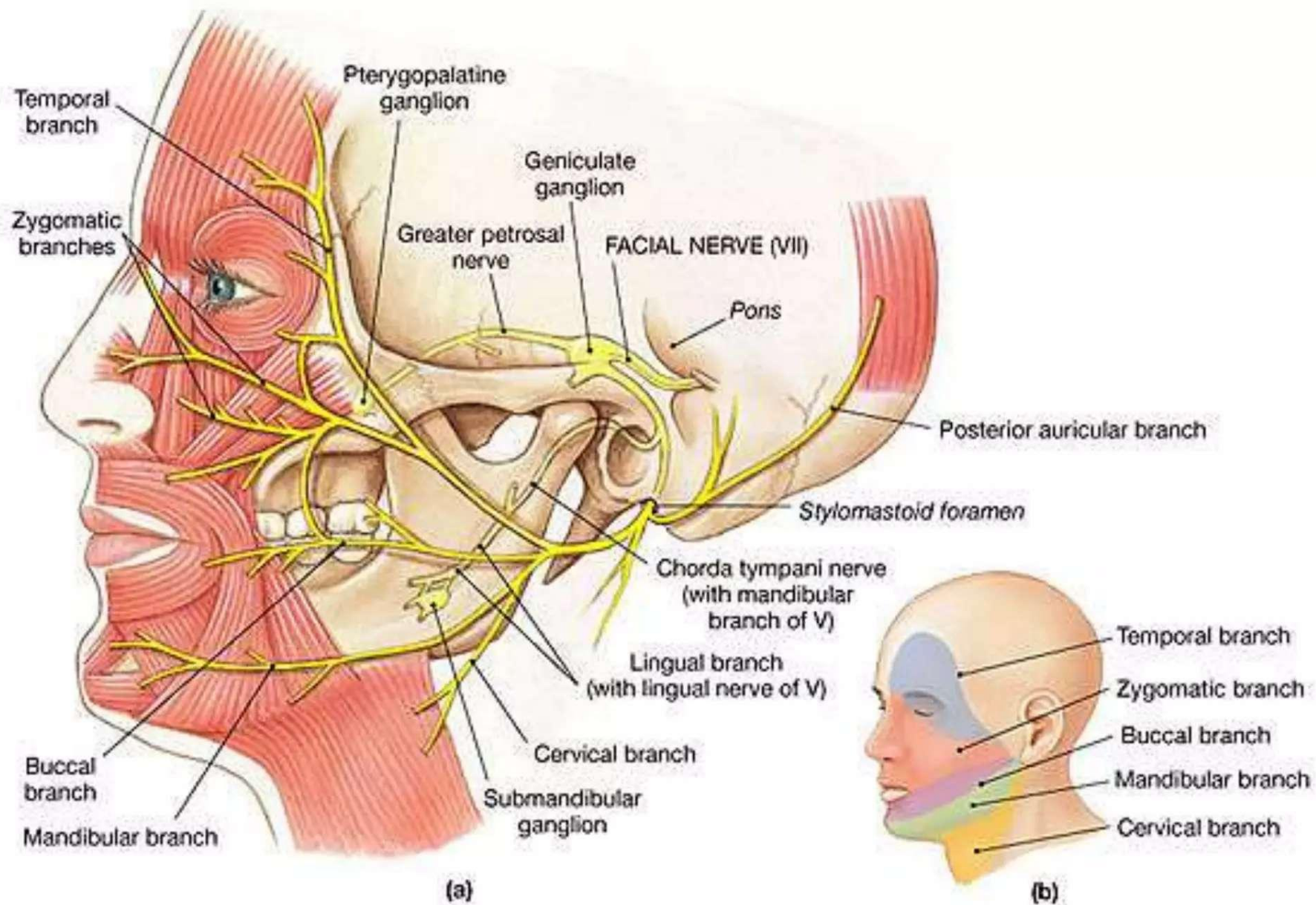
➤ **2 Nervus intermedius:**

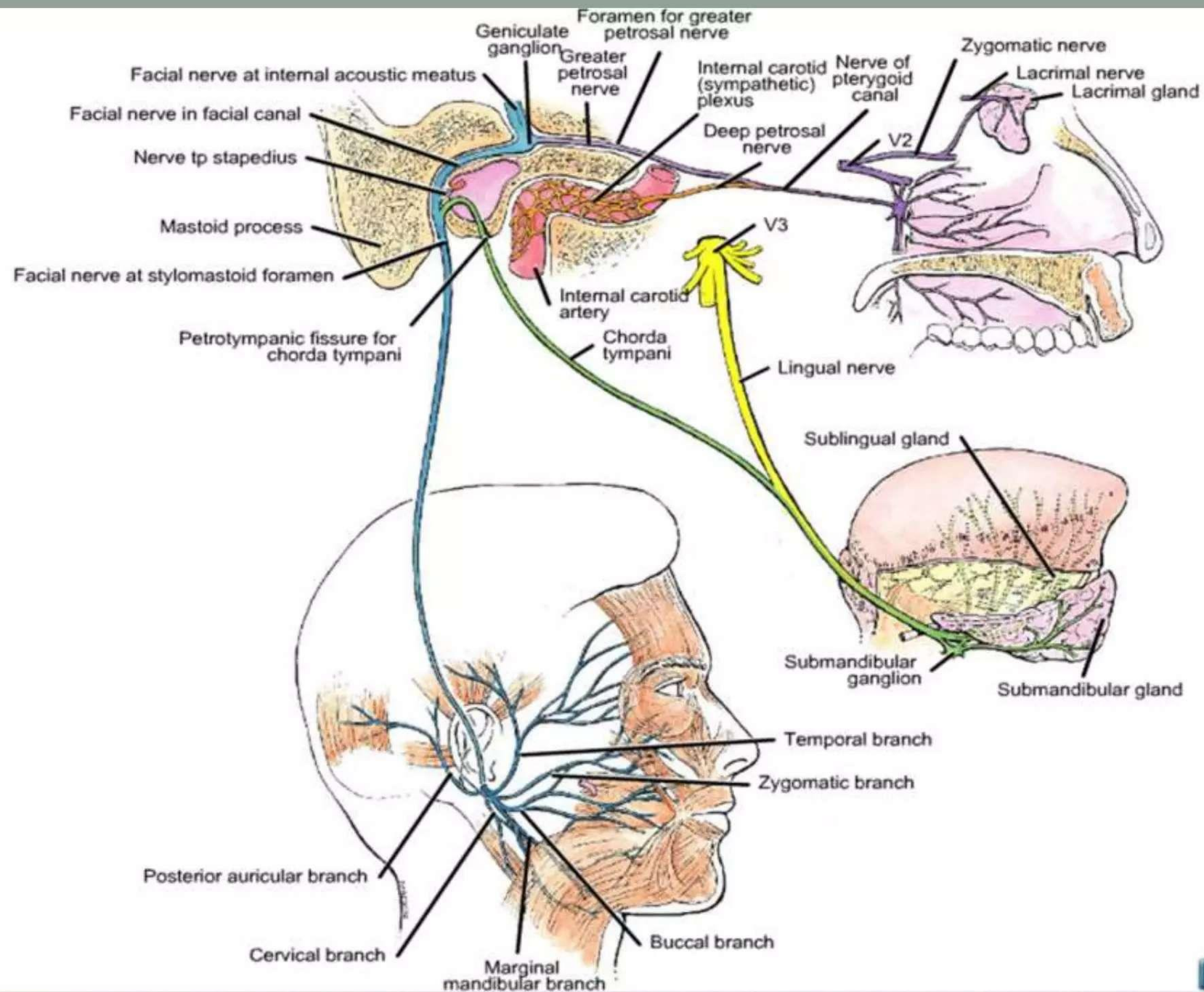
- Sensory and autonomic component of the facial nerve.
- Runs in a position intermediate between CNs VII and VIII across the CPA
- At first external genu, NI fuses with the geniculate ganglion.
- Sensory cells located in the geniculate ganglion are general somatic afferent (GSA) and special visceral afferent (SVA).

- GSA fibers carry exteroceptive impulses from region of the external auditory canal and tympanic membrane.
- SVA fibers convey taste from the anterior two-thirds of the tongue.
- Autonomic component of the NI consists of preganglionic general visceral efferent parasympathetic fibers from superior salivatory and lacrimal nuclei

Course and relations:

- Intracranial (intrapetrosal) course
- Extracranial course





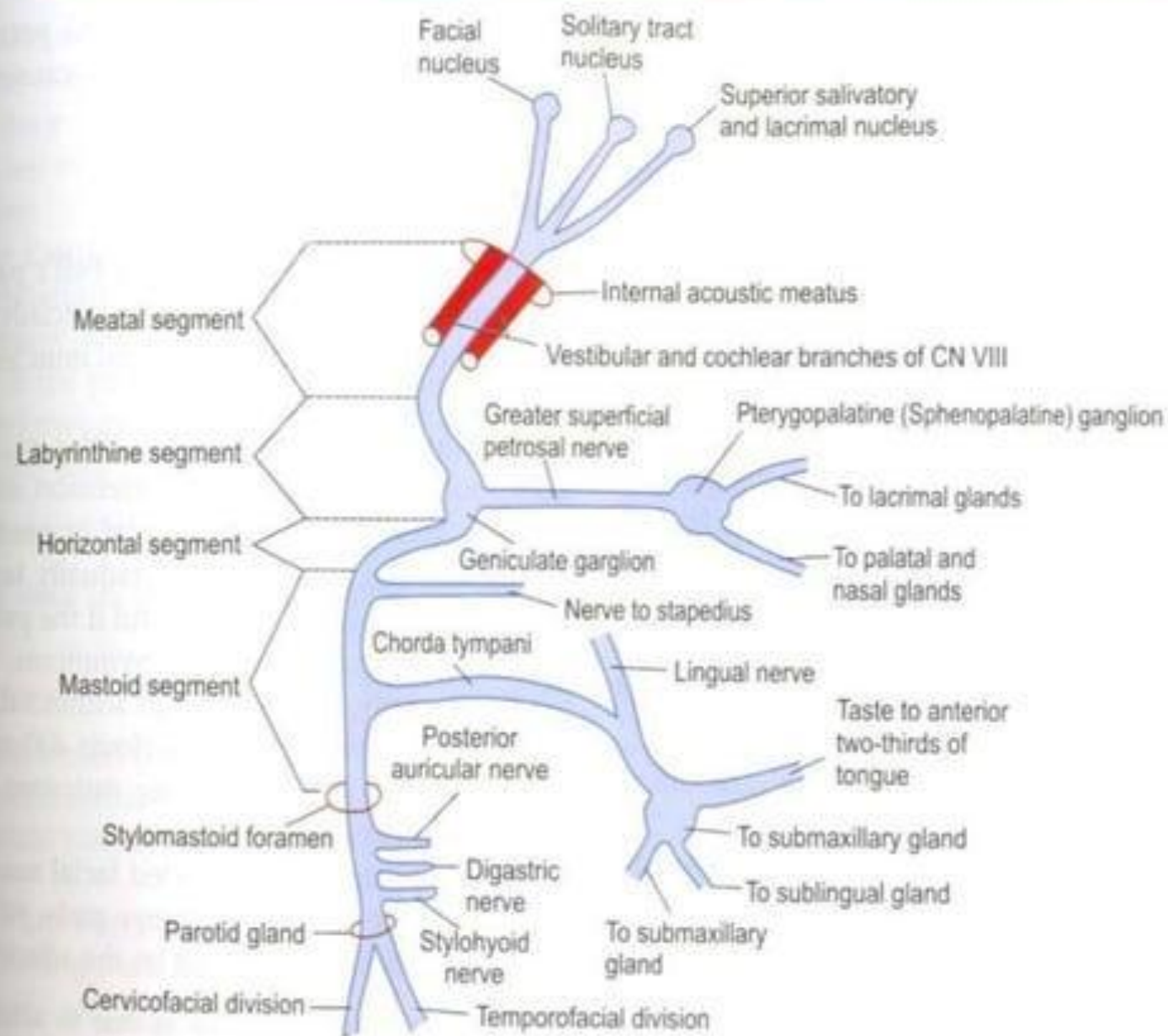


Fig. 8.34: The course of facial nerve

Facial Nerve: Functional Components

Branchial motor (special visceral efferent)	Supplies the muscles of facial expression; posterior belly of digastric muscle; stylohyoid, and stapedius.
Visceral motor (general visceral efferent)	Parasympathetic innervation of the lacrimal, submandibular, and sublingual glands, as well as mucous membranes of nasopharynx, hard and soft palate.
Special sensory (special afferent)	Taste sensation from the anterior 2/3 of tongue; hard and soft palates.
General sensory (general somatic afferent)	General sensation from the skin of the concha of the auricle and from a small area behind the ear.

Branches of Distribution

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graph TD; A[Branches of Distribution] --> B[Facial canal]; A --> C[Stylomastoid foramen]; A --> D[In face];
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Facial canal

- A. Nerve to stapedius
- B. Greater petrosal nerve
- C. Chorda tympani

Stylomastoid foramen

- A. Posterior auricular
- B. Nerve to stylohyoid
- C. Nerve to digastric (posterior belly)

In face

- A. Temporal
- B. Zygomatic
- C. Buccal
- D. Marginal mandibular
- E. Cervical

Within the facial canal:

- **1- Greater superficial petrosal nerve (GSPN) :** carries preganglionic parasympathetic fibers
 - These fibers are conveyed by the NI to geniculate ganglion.
 - Pass through the ganglion without synapsing into the greater petrosal nerve, which goes forward through the hiatus of the facial canal to join deep petrosal nerve from the carotid sympathetic plexus to form the vidian nerve, or the nerve of the pterygoid canal, which runs to the sphenopalatine ganglion, from where postganglionic fibers proceed to the lacrimal gland.

- **2-Nerve to stapedius:** supplies the stapedius muscle.

3- Chorda tympani nerve:

- It arises from the facial nerve 6 mm above the stylomastoid foramen and runs upwards to perforate the posterior bony wall of the tympanic cavity.
- It carries taste and general visceral afferent (GVA) fibers as well as preganglionic parasympathetics.

General Somatic afferents (GSA)/Sensory

- Fibers carrying somatosensory afferents in chorda tympani have their cell bodies in the geniculate ganglion.
- Peripheral processes innervate part of the external auditory canal, tympanic membrane, lateral surface of the pinna, and small area behind the ear and over the mastoid process.
- Their central processes terminate in the spinal tract and nucleus of the trigeminal

Special Visceral Afferent/Taste

- Taste sensation from the anterior two-thirds of tongue is carried through lingual nerve to the chorda tympani, then to geniculate ganglion.
- Also carry taste sensation from the mucosa of soft palate through the sphenopalatine ganglion.
- Central processes carrying taste and GVA sensation terminate in the nucleus of the solitary tract.

- Solitary tract sends communications to the superior and inferior salivatory nuclei, which send parasympathetics to salivary glands.
- Other fibers synapse in reticular formation; next order neurons form a component of the reticulospinal tract bilaterally to synapse with sympathetic neurons in the intermediolateral gray column of the upper thoracic spinal cord.
- These send sympathetic innervation via the superior cervical ganglion to the salivary glands.

➤ **General Visceral Efferent/Parasympathetic**

- Chorda tympani also carries Preganglionic parasympathetic fibers to the submandibular ganglion.
- Postganglionic fibers convey secretory and vasodilator impulses to the submandibular and sublingual salivary glands and mucous membranes of mouth and tongue.
- These glands also receive sympathetic innervation through superior cervical ganglion and the carotid plexus.
- Parasympathetic fibers cause vasodilation and a copious, thin, watery secretion high in enzymes;
- Sympathetic fibers cause vasoconstriction and a scant, thick, mucoid secretion low in enzyme content.

➤ **II- At exit from the stylomastoid foramen**

- **1- Posterior auricular nerve:**
Supplies the occipitalis , posterior auricular, and transverse and oblique auricular muscles.
- **2- Digastric branch:**
Posterior belly of digastric muscle
- **3- Stylohyoid branch:**
Stylohyoid muscle

TABLE 16.1**Muscles of the Face, Their Actions, and Innervations**

Nerve Branch	Muscle Innervated	Muscle Action
Temporal branch	Frontalis	Raises eyebrows and skin over the root of the nose; draws scalp forward, throwing forehead into transverse wrinkles
	Corrugator (corrugator supercilii)	Draws eyebrow down and medially, produces vertical wrinkles in the forehead (the frowning muscle)
	Upper part of the orbicularis oculi (orbicularis palpebrarum)	Eyelid sphincter; palpebral portion narrows palpebral fissure and gently closes eyelids; orbital portion draws skin of forehead, temple, and cheek toward medial orbit, pulls eyebrow down, draws skin of cheek up; closes eye firmly
	Occipitalis	Draws scalp backward
Zygomatic Buccal	Procerus (pyramidalis nasi)	Draws medial eyebrow downward, produces transverse wrinkles over bridge of nose
	Lower and lateral orbicularis oculi	Eyelid sphincter
	Orbicularis oculi	Eyelid sphincter
	Buccinator	Compresses cheeks, keeps food under pressure of cheeks in chewing
	Zygomaticus	Draws mouth backward and upward
	Nasalis (compressor nares)	Depresses cartilaginous portion of nose, draws the ala toward septum
	Levator anguli oris (caninus)	Raises angle of mouth
	Levator labii superioris (quadratus labii superioris)	Elevates upper lip, dilates nostril
Mandibular	Lower part of the orbicularis oris	Sphincter of the mouth; closes lips; superficial fibers protrude lips; deep fibers draw lips in and press them against teeth
	Mentalis	Protrudes lower lip, wrinkles skin of chin
	Risorius	Retracts angle of mouth
	Triangularis (depressor anguli oris)	Depresses angle of mouth
	Depressor labii inferioris (quadratus labii inferioris)	Draws lower lip downward and lateralward
Cervical	Platysma	Pulls lower lip and angle of mouth down; depresses lower jaw; raises and wrinkles skin of neck

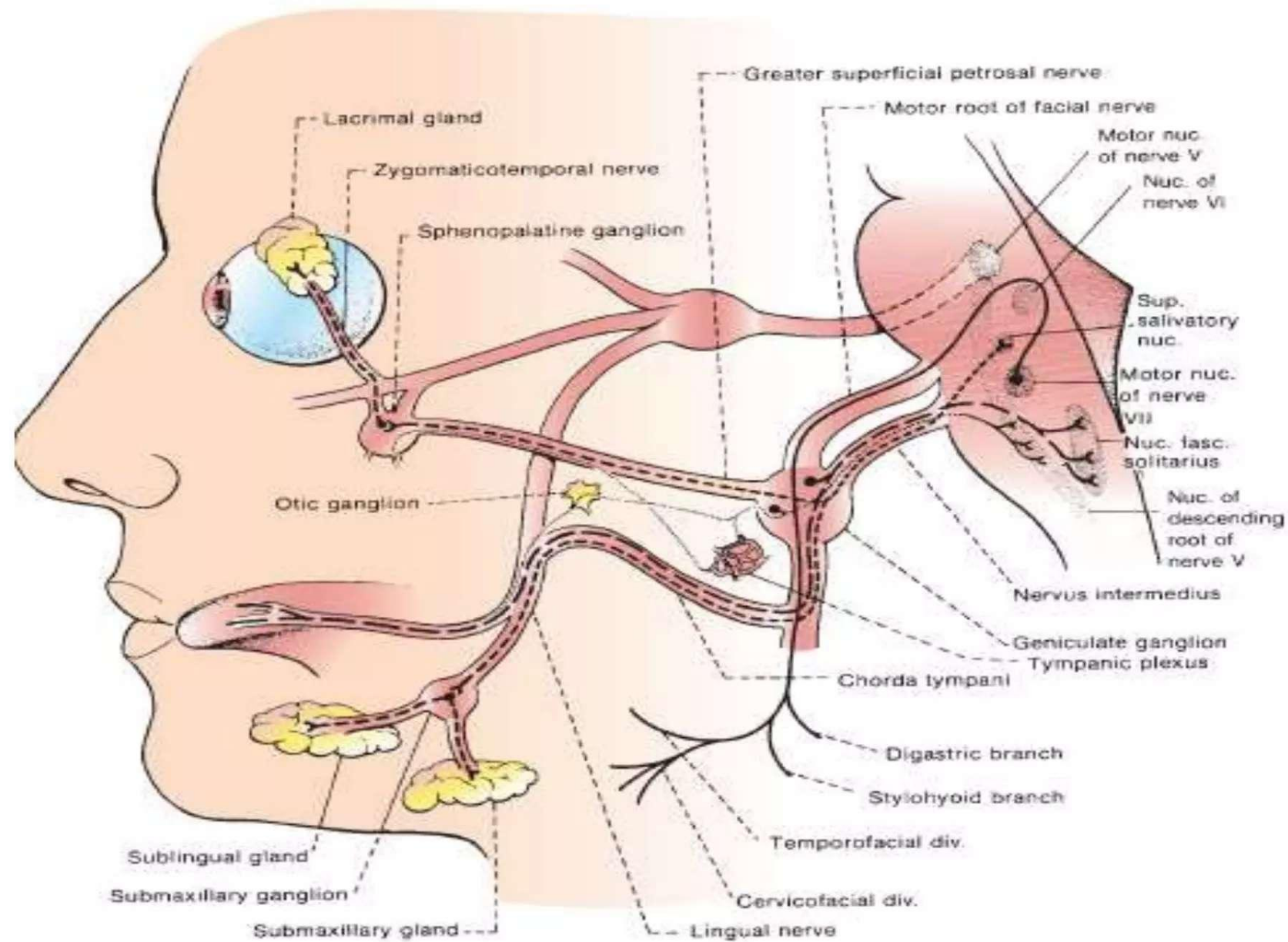


FIGURE 16.3 Course and branches of the facial nerve.

GANGLIA ASSOCIATED WITH FACIAL NERVE

- Geniculate ganglion
- Submandibular ganglion
- Pterygopalatine ganglion

Geniculate Ganglion

- The geniculate ganglion (from Latin genu, for "knee") is an L-shaped collection of fibers and sensory neurons of the facial nerve located in the facial canal of the head.
- It receives fibers from the motor, sensory, and parasympathetic components of the facial nerve and sends fibers that will innervate the lacrimal glands, submandibular glands, sublingual glands, tongue, palate, pharynx, external auditory meatus, stapedius, posterior belly of the digastric muscle, stylohyoid muscle, and muscles of facial expression.

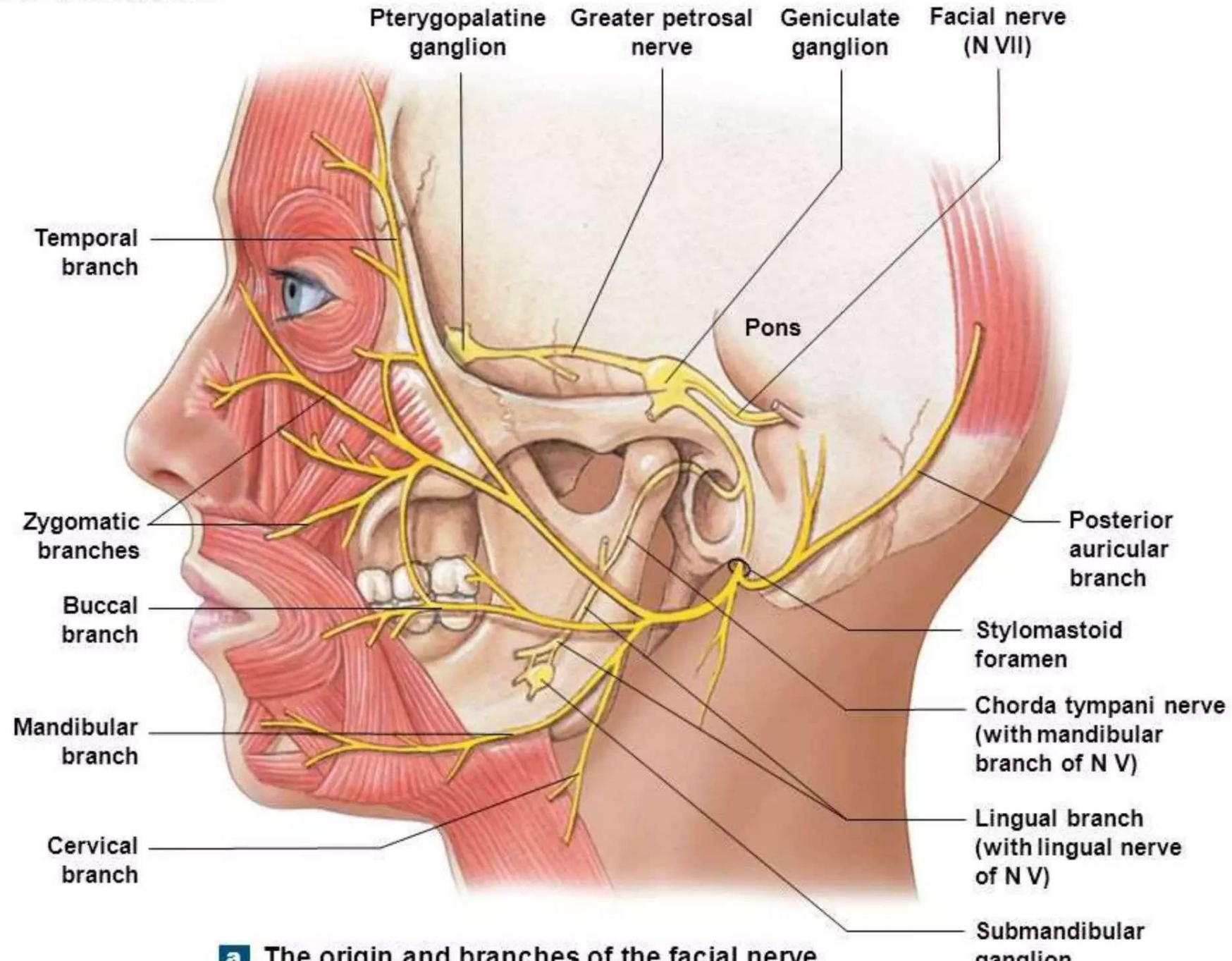
Submandibular Ganglion

- The submandibular ganglion is small and fusiform in shape. It is situated above the deep portion of the submandibular gland, on the hyoglossus muscle, near the posterior border of the mylohyoid muscle.
- The ganglion 'hangs' by two nerve filaments from the lower border of the lingual nerve (itself a branch of the mandibular nerve, CN V3). It is suspended from the lingual nerve by two filaments, one anterior and one posterior. Through the posterior of these it receives a branch from the chorda tympani nerve which runs in the sheath of the lingual nerve.

Pterygopalatine Ganglion

- The pterygopalatine ganglion (meckel's ganglion, nasal ganglion or sphenopalatine ganglion) is a parasympathetic ganglion found in the pterygopalatine fossa.
- It's largely innervated by the greater petrosal nerve (a branch of the facial nerve); and its axons project to the lacrimal glands and nasal mucosa

Figure 14-24a The Facial Nerve.



a The origin and branches of the facial nerve

CLINICAL EXAMINATION

➤ Examination of the Motor Functions

➤ Inspection-

- Facial asymmetry, nasolabial fold with forehead wrinkles, movements during spontaneous facial expression
- Tone of the muscles of facial expression,
- Atrophy and fasciculations
- Abnormal muscle contractions and involuntary movements
- Spontaneous blinking for frequency and symmetry.

Testing of Facial Nerve Branches

- **Testing the temporal branches of the facial nerve** – patient is asked to frown and wrinkle his or her forehead.
- **Testing the Zygomatic branches of the facial nerve** patient is asked to close their eyes tightly
- **Testing the buccal branches of the facial nerve**
 - Puff up cheeks (buccinator)
 - Smile and show teeth (orbicularis oris)
 - Tap with finger over each cheek to detect ease of air expulsion on the affected side

Examination of Sensory Functions

- Hypesthesia of posterior wall of the external auditory meatus in proximal facial nerve lesions.
- Taste on anterior two-thirds of the tongue-use four substances for testing:
 - Sucrose (sweet), sodium chloride (salty), quinine (bitter), and citric acid (sour).
 - Patient with a peripheral pattern of facial weakness has impaired taste, the lesion is proximal to the junction with the chorda tympani.

Examination of Secretory Functions

- Tear production may be quantitated with the Schirmer test.
- Lacrimal reflex is tearing, usually bilateral, caused by stimulating the cornea.
- Nasolacrimal reflex is elicited by mechanical stimulation of the nasal mucosa, or by chemical stimulation using irritating substances such as ammonia.
- Abnormalities of salivation are usually suggested by the history.

Examination of facial nerve

- *Show the teeth*
- *Open his mouth—compare nasolabial folds*
- *Close his eyes*
- *Frown*
- *Wrinkle forehead*
- *Raise eyebrows*
- *Bare his teeth and open his mouth*
- *Blowing out cheeks*
- *Pursuing the lips –strength and weakness*

Clinically

- Corner of mouth droops
- Crease and skin folds effaces
- Forehead is unfurrowed
- Eyelids will not close
- Eye on the paralysed side rolls upward –
BELL'S PHENOMENON
- Lower lid sags and falls away from conjunctiva
- Tears spill over cheek
- Food collects between the teeth and lips
- Saliva may dribble from the corner of the mouth
- Heaviness or numbness of the face
- Sensory loss rarely demonstrable

LMN FACIAL PALSY



Fig. 10.2 Lower motor neuron facial paralysis. Both upper and lower parts of the face are equally involved. Note absence of wrinkling of right forehead and visible sclera on screwing up eyes.

Facial Weakness

- Two types of neurogenic facial nerve weakness:
 - **Peripheral or lower motor neuron** - result from a lesion anywhere from the CN VII nucleus in the pons to the terminal branches in the face.
 - **Central facial palsy (CFP)** - due to a lesion involving the supranuclear pathways before they synapse on the facial nucleus.

Peripheral Facial Palsy

- There is flaccid weakness of all the muscles of facial expression on the involved side, both upper and lower face, and the paralysis is usually complete



- Palpebral fissure is open wider than normal, and there may be inability to close the eye (lagophthalmos).
- Very mild PFP may produce only slower and less complete blink on the involved side.
- **Bell's phenomenon-** Attempting to close involved eye causes a reflex upturning of the eyeball
- **Levator sign of Dutemps and Céstan-** Patient look down, then close the eyes slowly; because the function of levator palpebrae superioris is no longer counteracted by orbicularis oculi, upper lid on the paralyzed side moves upward slightly.

- **Negro's sign-** eyeball on the paralyzed side deviates outward and elevates more than the normal one when the patient raises her eyes.
- **Bergara-Wartenberg sign-** loss of the fine vibrations palpable with the thumbs or fingertips resting lightly on the lids as the patient tries to close the eyes as tightly as possible.
- **Platysma sign of Babinski-** asymmetric contraction of the platysma, less on the involved side, when the mouth is opened

Bell's Palsy

- Most common form of lower motor neuron facial palsy.
- Incidence is 23/1,00,000
- 1 in 60 life time occurrence of single episode
- Affects men and women equally , all ages ,all times of the year.
- Increased occurrence in the elderly diabetics, hypertensives than in the common people.
- Increased incidence in women during the third trimester of pregnancy 2 weeks preceding delivery ,first two weeks postpartum.

➤ **Etiology:**

- Idiopathic
- Herpes simplex virus 1
- Herpes zoster is probably second most common viral infection associated with PFP.
- Other viruses implicated include cytomegalovirus, Epstein-Barr virus, human herpes virus 6, and coxsackie.
- Inactivated intra nasal influenza vaccine

Clinical features

- Onset of bell's palsy is acute.
- $\frac{1}{2}$ of the cases attain maximum paralysis in 48 hours.
- All cases are clinically prominent by 5 days.
- Pain behind the ear may precede the paralysis by a day or two .
- Impairment of taste is present to some degree in all cases – rarely beyond second week of paralysis.
- Hyperacusis or distortion of sound in ipsilateral ear ---paralysis of stapedius muscle.
- Paralysis is partial in 30%,complete in 70%cases.
- About 1% of cases are bilateral

- Enhancement of the facial nerve on gadolinium enhanced MRI
- Increased lymphocytes ,mononuclear cells in CSF.

➤ **Prognosis**

- 80% patients recover within a few weeks.2-12 weeks.
- 10%--permanent long term sequelae.
- 8%--recurrence

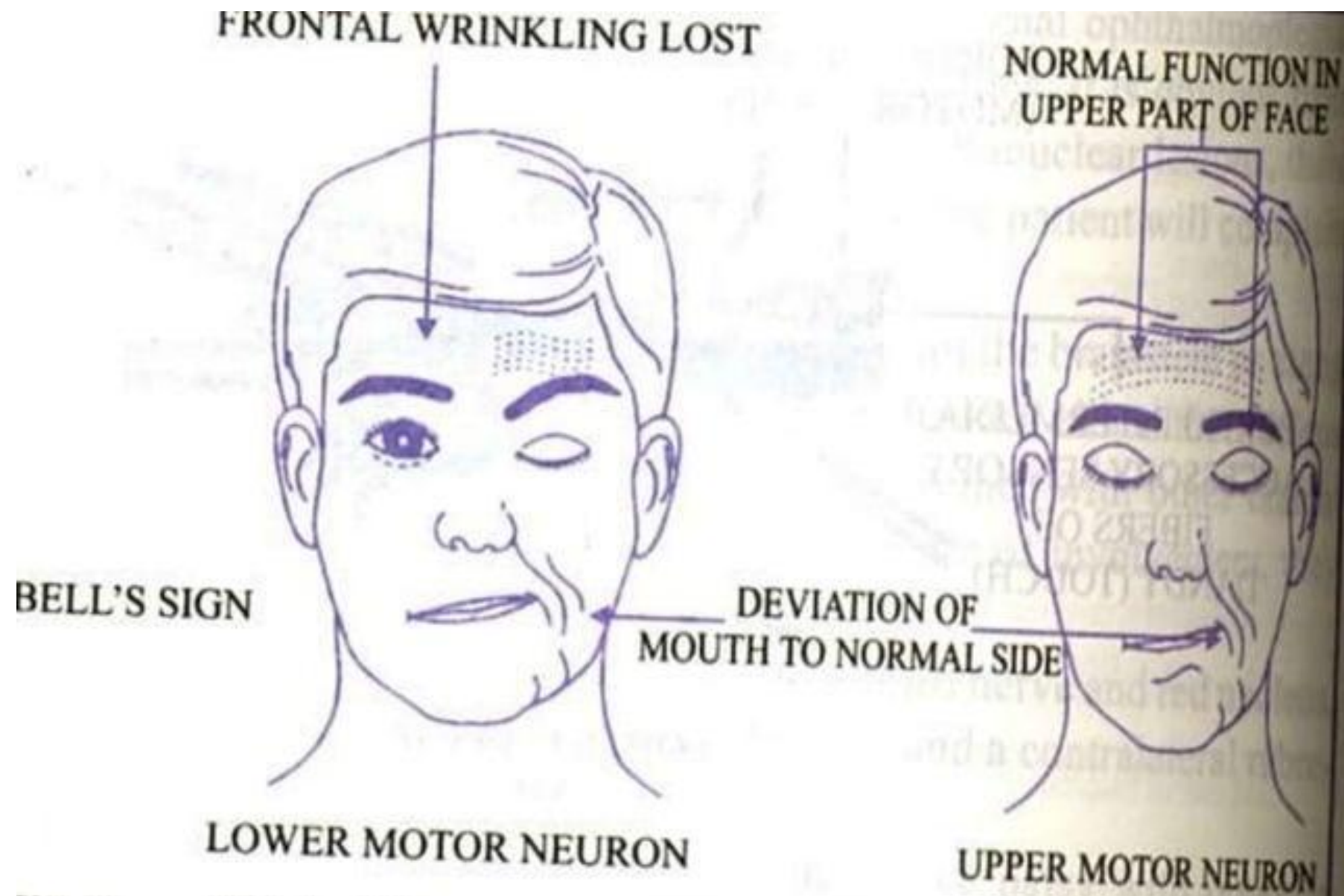


Fig. 5 Clinical features of upper and lower motor neurone facial paralysis